

USTHB University
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جامعة هواري بومدين

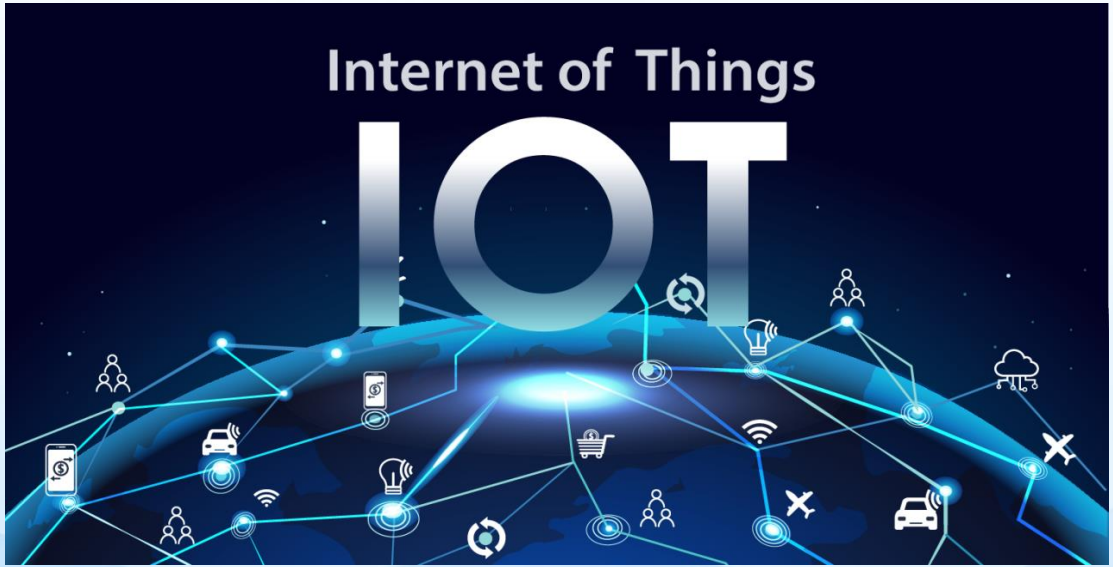
Internet of Things (IoT)

02 IoT Components and Sensors



5th Engineers | AMEUR IMANE | 2026/2027

1. IoT Components and Sensors



IoT Devices

1

Simple Sensors
Temperature, humidity, motion

2

Smart Devices
Watches, thermostats, cameras

3

Actuators
Relays, valves, motors, locks

4

Gateways
Connect devices to the Internet



What is IoT?

Connecting the physical world to digital services

Environment
temperature, motion

→

Sensor
collects data

→

Controller
processes data

→

Network
transmits data

→

Cloud
services

Smart Home

Thermostats, lighting, security systems

Smart Industry

Machine monitoring and automation

Smart Agriculture

Soil, irrigation, climate monitoring



An IoT Node: How the Parts Fit Together

Sensors

- Measure the physical environment.
- Temperature
- Humidity
- Light
- Motion

→

Microcontroller

- Reads inputs, processes data and makes decisions.
- CPU
- Memory
- GPIO
- ADC

→

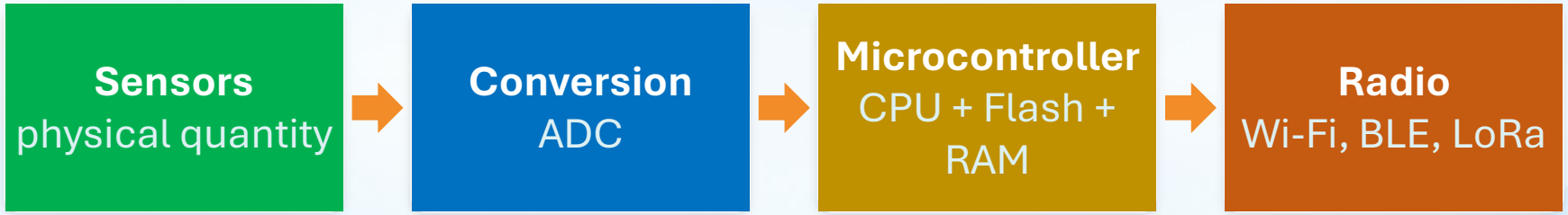
Actuators

- Act on the physical world
- Motor
- Relay
- Pump
- Buzzer

Connectivity (Wi-Fi, BLE, LoRa, Zigbee…) links the node to the gateway and the cloud

Power supply (battery, mains, solar) feeds every block.

Typical IoT Device Architecture



Actuators (via driver)

Power supply and energy management (battery, mains, solar)

Interfaces: GPIO, I2C, SPI, UART

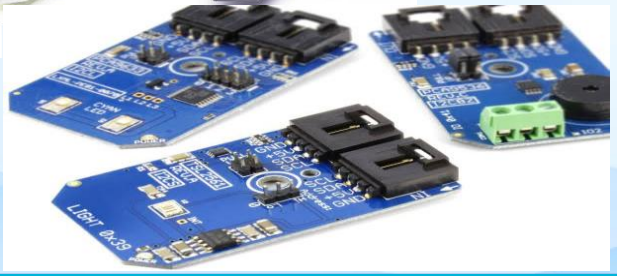
- The microcontroller runs the firmware and drives all the blocks

Sensors

Sensors are the input layer of an IoT system. They measure physical conditions and convert them into electrical or digital data that a microcontroller can process.







Sensors: From Physical Quantity to Digital Data

A sensor converts a physical quantity (temperature, light, motion…) into an electrical signal.



- Temperature
- Humidity
- Pressure
- Light
- Motion
- Distance
- Gas concentration

Analog sensors output a voltage (e.g. LM35, LDR) and need an ADC.

Digital sensors output data over I^2C, SPI, or 1-Wire (e.g. DHT22, BME280).

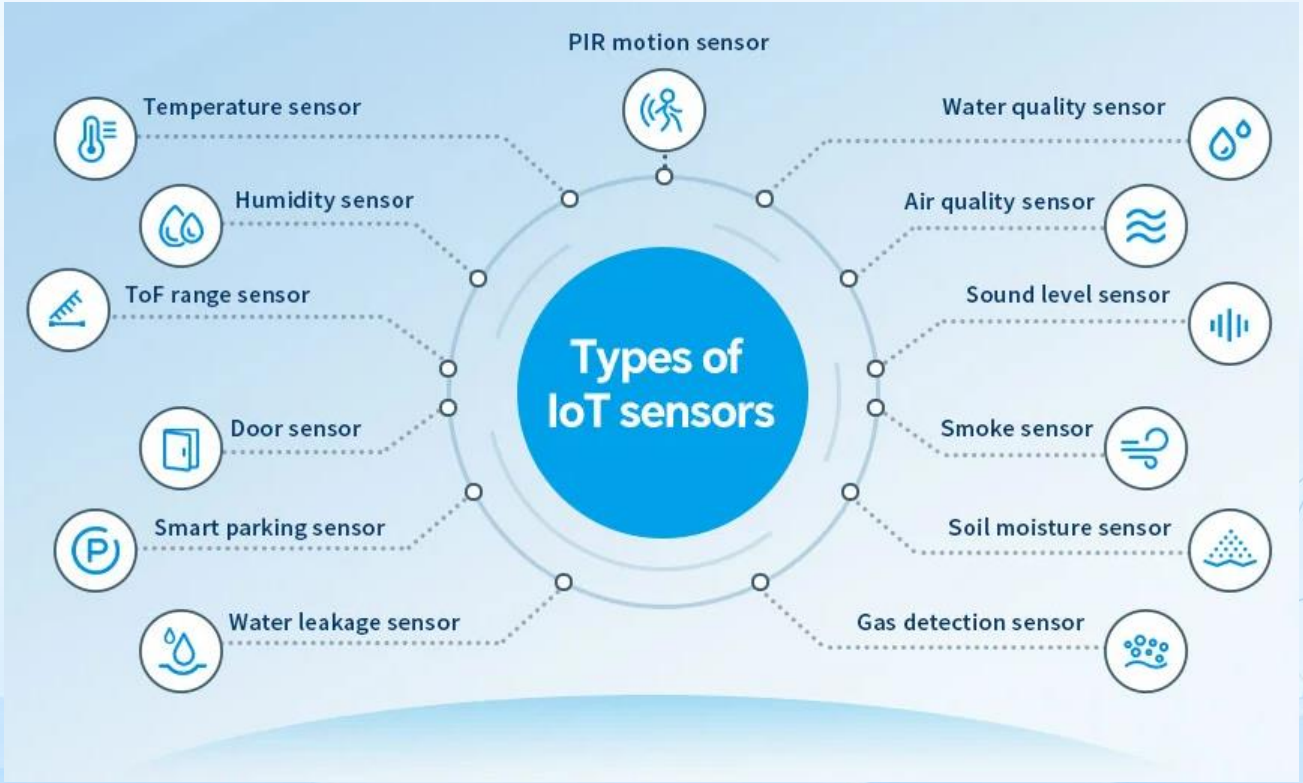
Common Types of Sensors

Measured quantity	Sensor	Typical use
Temperature / humidity	DHT22, BME280, LM35	Smart home, greenhouse
Light	LDR, BH1750	Street lighting, displays
Motion / presence	PIR, accelerometer (MPU6050)	Alarms, wearables
Distance	Ultrasonic HC-SR04, LiDAR	Parking, robotics
Gas / air quality	MQ-2, MQ-135, SGP30	Safety, smart city
Soil moisture	Capacitive soil sensor	Smart irrigation
Position	GPS (NEO-6M)	Asset tracking



Common Types of Sensors





Analog vs Digital Sensors

Two common ways of representing sensor information

Analog Sensors

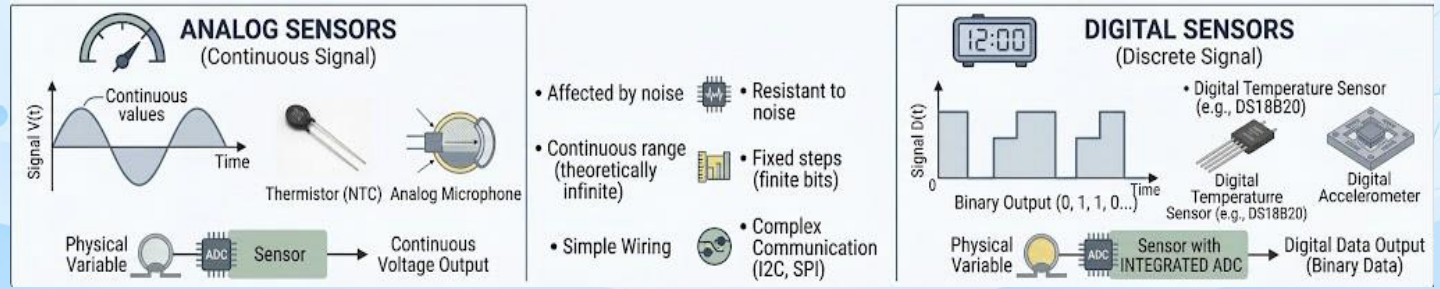
Analog Sensors are shown like a dimmer switch or speedometer they produce a smooth, continuous flow of signals.

- Continuous electrical output
- Example: 0 to 3.3 V
- Require ADC conversion
- Examples: LDR, soil moisture, gas sensors

Digital Sensors

Digital Sensors are shown like an on/off switch they convert signals into discrete steps and binary data (1s and 0s).

- Discrete digital data
- Often include internal processing
- Communicate using I^2C, SPI, UART or OneWire
- Examples: DHT22, DS18B20



Key Sensor Characteristics

Range

Min–max values it can measure



Sensitivity

Output change per unit of input



Accuracy

Closeness to the true value



Resolution

Smallest detectable change



Response time

Delay to reach a stable reading



Power use

Critical for battery nodes



Tip: accuracy ≠ precision. A sensor can be repeatable yet consistently wrong (needs calibration).

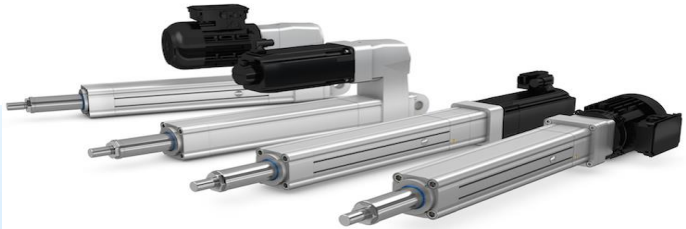
Key sensor characteristics

Characteristic	Definition & Measurement	Practical Significance
Range	Min–max values the sensor can measure	Defines the upper and lower operational limits of the device in target environments.
Sensitivity	Output change per unit of input	Determines how responsive the sensor is to subtle fluctuations in the physical variable.
Accuracy	Closeness to the true value	Minimizes measurement error and ensures high data fidelity and reliability.
Resolution	Smallest detectable change	Dictates the fine-grained detail and quantization levels of the captured signal.
Response Time	Delay to reach a stable reading	Crucial for real-time monitoring, fast-moving phenomena, and closed-loop control systems.
Power Use	Energy consumption profile	Critical for battery-powered IoT nodes, edge devices, and long-term unattended deployments.

Actuators



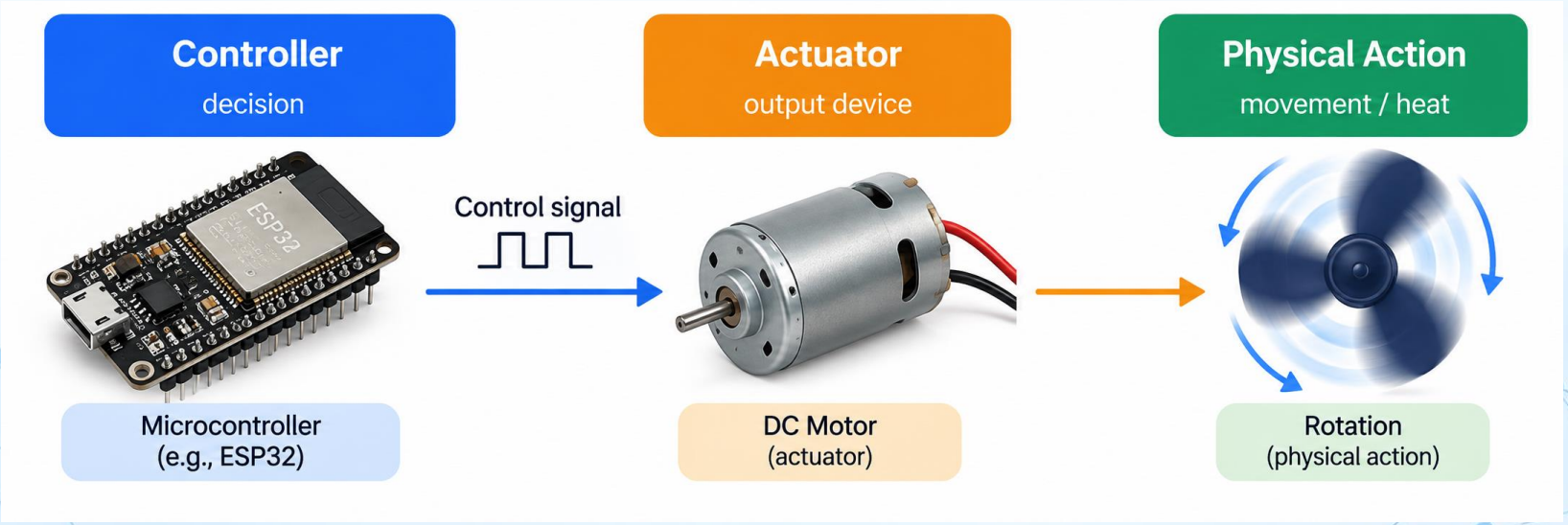
Actuators are the output layer of an IoT system. They convert control commands into physical effects such as motion, light, sound, switching, pumping, or flow control.





Actuators

Actuators convert control signals into physical actions



- LED: produces light
- Motor: produces rotation
- Relay: switches a load
- Buzzer: produces sound
- Pump: moves liquid
- Valve: controls flow

Actuators: Turning Decisions into Action

Actuator	What it does	Control
 LED / buzzer	Light or sound indication	GPIO pin (+ resistor)
 Relay	Switches mains-powered loads (lamp, pump)	GPIO + transistor driver
 DC motor	Continuous rotation (fan, wheel)	PWM + H-bridge (L298N)
 Servo motor	Precise angle (0–180°)	PWM signal
 Stepper motor	Exact steps (3D printer)	Driver (A4988)
 Solenoid valve	Opens or closes water/air flow	Relay or MOSFET

Never drive motors or relays directly from a GPIO pin: use a driver and protect with a flyback diode.

Sensors vs Actuators

Input versus output in an IoT system

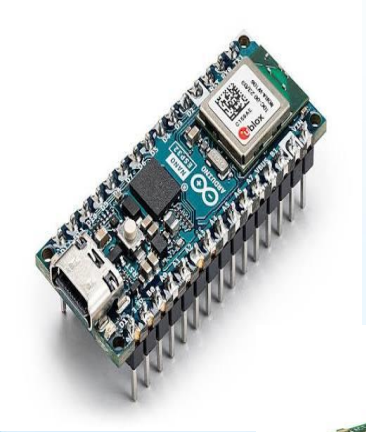
Aspect	Sensor	Actuator
Role	Input device	Output device
Function	Measures environment	Changes environment
Direction	Sends data to controller	Receives commands
Example	Temperature sensor	**Fan** / motor / relay

SENSOR = SENSE

ACTUATOR = ACT

Microcontrollers

Microcontrollers are compact embedded computers that read sensors, process data, execute control logic, communicate with networks, and control actuators in real time.





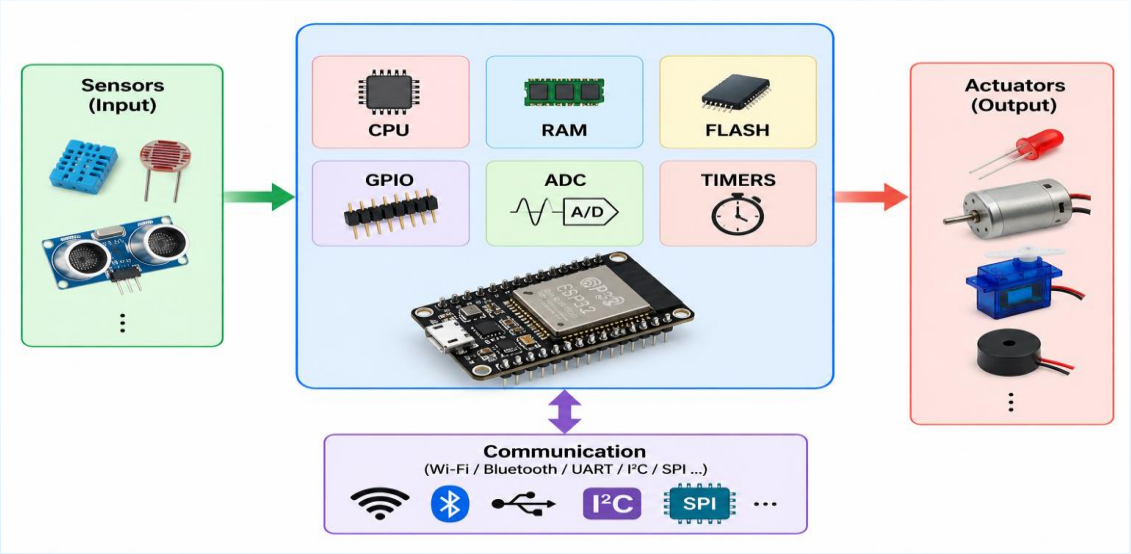


Microcontrollers

A microcontroller (MCU) is a small computer on a single chip, designed for low-power, real-time control.

Typical tasks

- Read sensors
- Process data
- Make decisions
- Control actuators
- Communicate with networks



- Low power
- Low cost
- Real-time control
- Compact hardware
- Embedded software

Popular IoT choices: ESP32, ESP8266, Arduino-class boards, Raspberry Pi Pico W.

The embedded “brain” of many IoT devices

Popular IoT Microcontrollers

Common platforms for prototyping and embedded systems

Arduino Uno

- Beginner-friendly
- ATmega328P
- No integrated Wi-



ESP8266

- Low cost
- Integrated Wi-Fi
- Popular for simple connected devices



ESP32

- Wi-Fi + Bluetooth
- More processing



Raspberry Pi Pico W

- RP2040 MCU
- Low cost
- Wi-Fi on Pico W



Choosing a Board

** A microprocessor board (full OS), not a microcontroller. Often used as the IoT gateway.*

Board	Core	Connectivity	Best for
Arduino Uno	ATmega328P, 8-bit, 16 MHz	None (add modules)	Learning, simple prototypes
ESP32	Dual-core 32-bit, up to 240 MHz	Wi-Fi + Bluetooth	Most IoT projects
Raspberry Pi Pico W	RP2040, dual-core, 133 MHz	Wi-Fi	Low-cost, MicroPython
STM32 (Nucleo)	ARM Cortex-M, 32-bit	Via modules	Industrial, low power
Raspberry Pi 4*	Quad-core Linux computer	Wi-Fi, Ethernet, BT	Gateways, vision, edge AI

GPIO, ADC and PWM

Three interfaces used constantly in embedded IoT systems.

Three essential microcontroller functions

GPIO

- General-Purpose Input/Output
- Connects digital sensors and actuators
- Can be configured as input or output



Digital input or output

ADC

- Analog-to-Digital Converter
- Converts sensor voltage into numbers
- Example: 0–3.3 V → digital value



Analog voltage → digital value

PWM

- Pulse Width Modulation
- Controls average power
- Used for LED brightness and motor speed



Pulse width controls average power

Microcontroller Interfaces

Interface	Use	Example
GPIO	Digital input/output	LED, button
ADC	Analog reading	Soil moisture sensor
I2C / SPI	Bus for sensors and displays	BME280 sensor
UART	Serial link	GPS module

Common Hardware Platforms

Board	Type	Strengths
Arduino Uno	8-bit microcontroller	Simple, ideal for learning
ESP32	Microcontroller + Wi-Fi/BLE	Inexpensive, very widespread
Raspberry Pi Pico	Microcontroller	MicroPython, low cost
Raspberry Pi	Linux mini-computer	Gateway and prototype

Microcontroller: all-in-one, low energy. Microprocessor: more powerful, full operating system



Connectivity

Connectivity links IoT devices to sensors, gateways, other devices, and cloud services. The best technology depends on range, bandwidth, power consumption, and network requirements.



Sensor Communication Protocols

How a microcontroller exchanges data with peripherals.

I^2C

2 wires

SDA + SCL



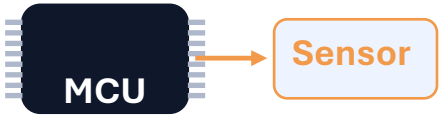
Multiple devices can share the bus.

- Two main lines: SDA and SCL
- Multiple devices share one bus
- Common for sensors and displays

SPI

Fast bus

MOSI, MISO, SCLK, CS



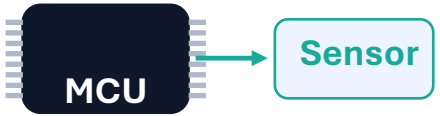
High speed, more wires.

- High-speed synchronous bus
- MOSI, MISO, SCLK and CS
- Common for displays and SD cards

UART

Serial link

TX + RX



Simple point-to-point communication.

- Serial communication
- TX = transmit
- RX = receive
- Common for GPS and GSM modules



IoT Connectivity Technologies

Choose connectivity according to range, power and bandwidth

Wi-Fi

- High bandwidth
- Home / buildings

Bluetooth LE

- Short range, low power
- Wearables

Zigbee

- Low-power mesh
- Smart home

LoRa

- Long range, low data rate
- Remote sensors

NB-IoT

- Cellular, low power
- Utility metering

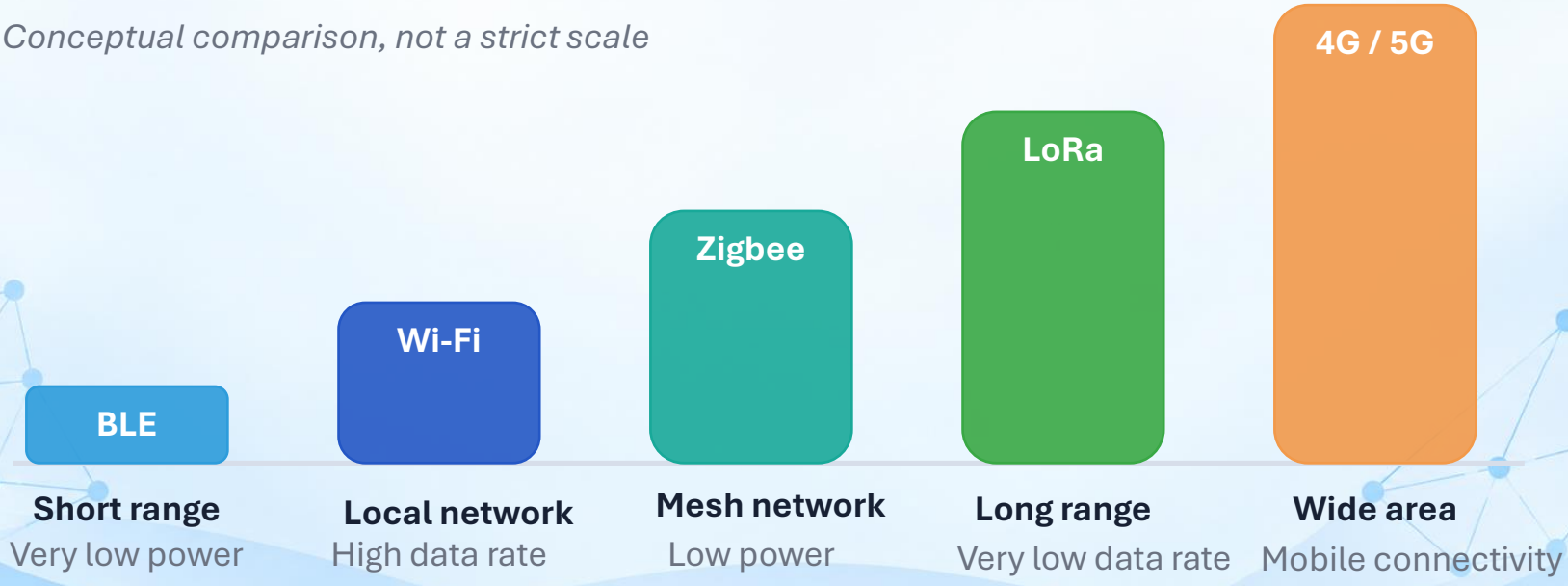
4G / 5G

- Wide-area, high bandwidth
- Vehicles / video

Wireless Connectivity for IoT

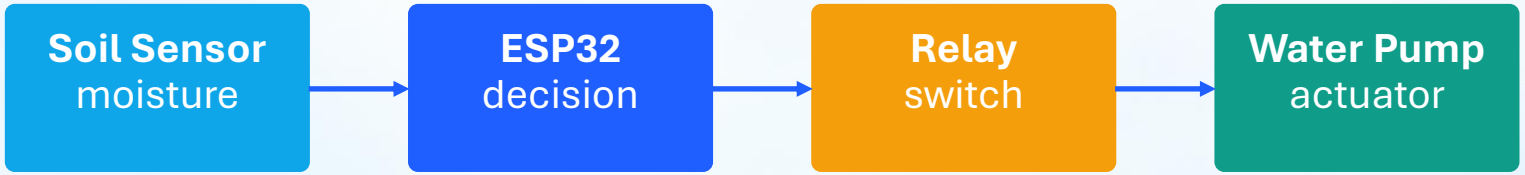
Different technologies optimize range, bandwidth, and power consumption.

Conceptual comparison, not a strict scale



Example: Smart Irrigation

A sensor-driven control system for agriculture




```
IF soil moisture < 30%  →  Pump ON
ELSE  →  Pump OFF
```

Case Study: Smart Irrigation

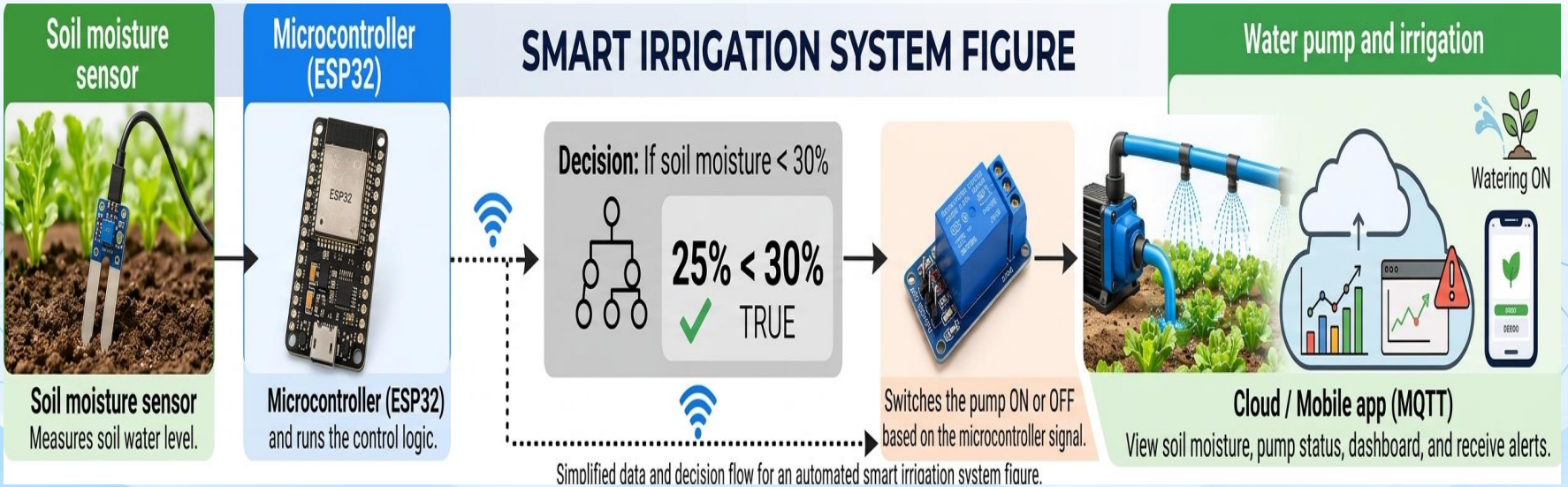
Sensor	Soil moisture + DHT22
MCU	ESP32 reads values every 10 min
Logic	If moisture < 30% → water
Actuator	Relay opens pump / valve
Cloud	MQTT → dashboard & alerts

```
IF soil moisture <
  30%   →   Pump ON
ELSE    →   Pump OFF
```

Example: Smart Irrigation

A sensor-driven control system for agriculture

IF soil moisture < 30% → Pump ON
ELSE → Pump OFF



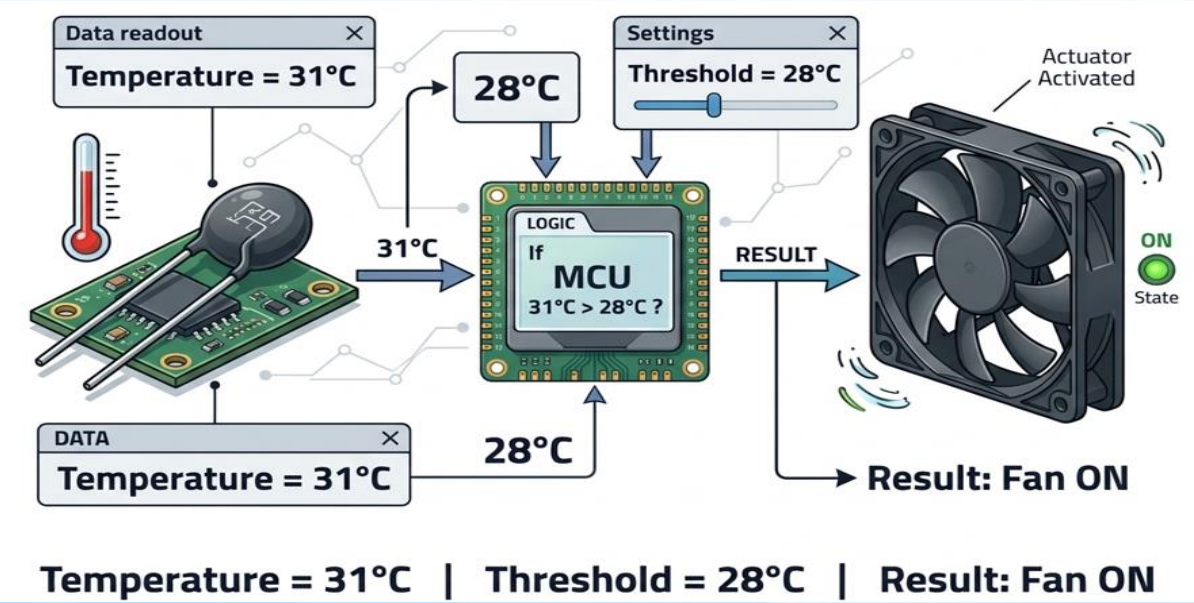
Simplified data and decision flow for an automated smart irrigation system figure.

https://www.youtube.com/watch?v=XTTsAl_Hk9o

Example: Smart Temperature Control

From sensing to automated actuation

DHT22 31°C → **ESP32** compare → **Decision** 31 > 28 → **Relay** ON → **Fan** running



https://www.youtube.com/shorts/tMy3JTRgxek

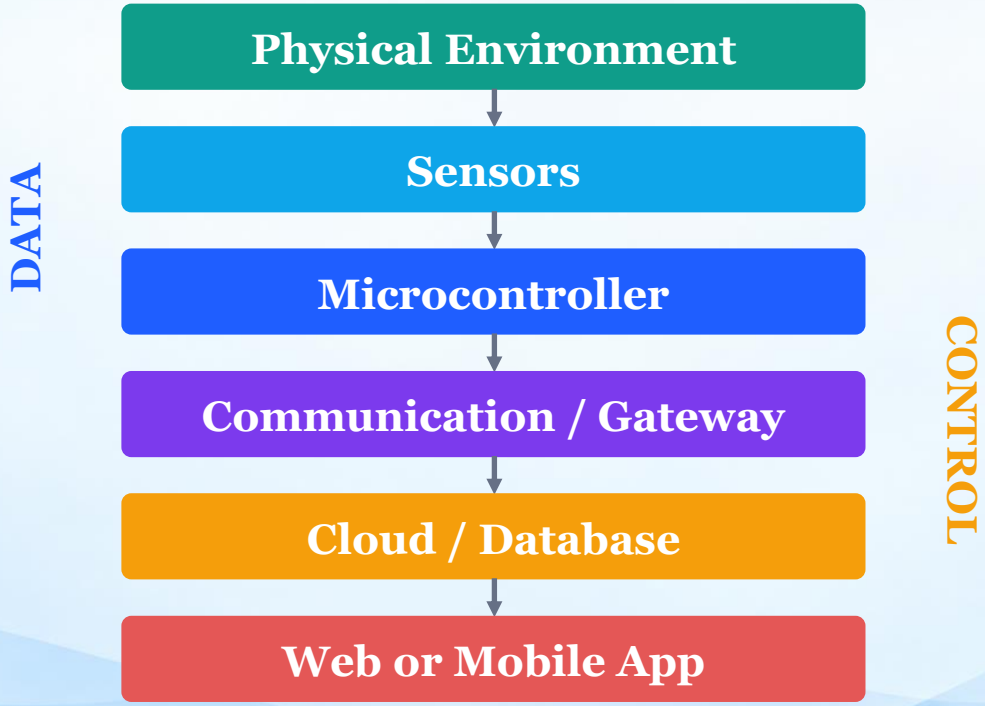
https://www.youtube.com/shorts/Uh_S9jFbujk

https://www.youtube.com/watch?v=Vq6aPo1krZI

The same ESP32 can also send the measurement to a cloud platform over Wi-Fi.

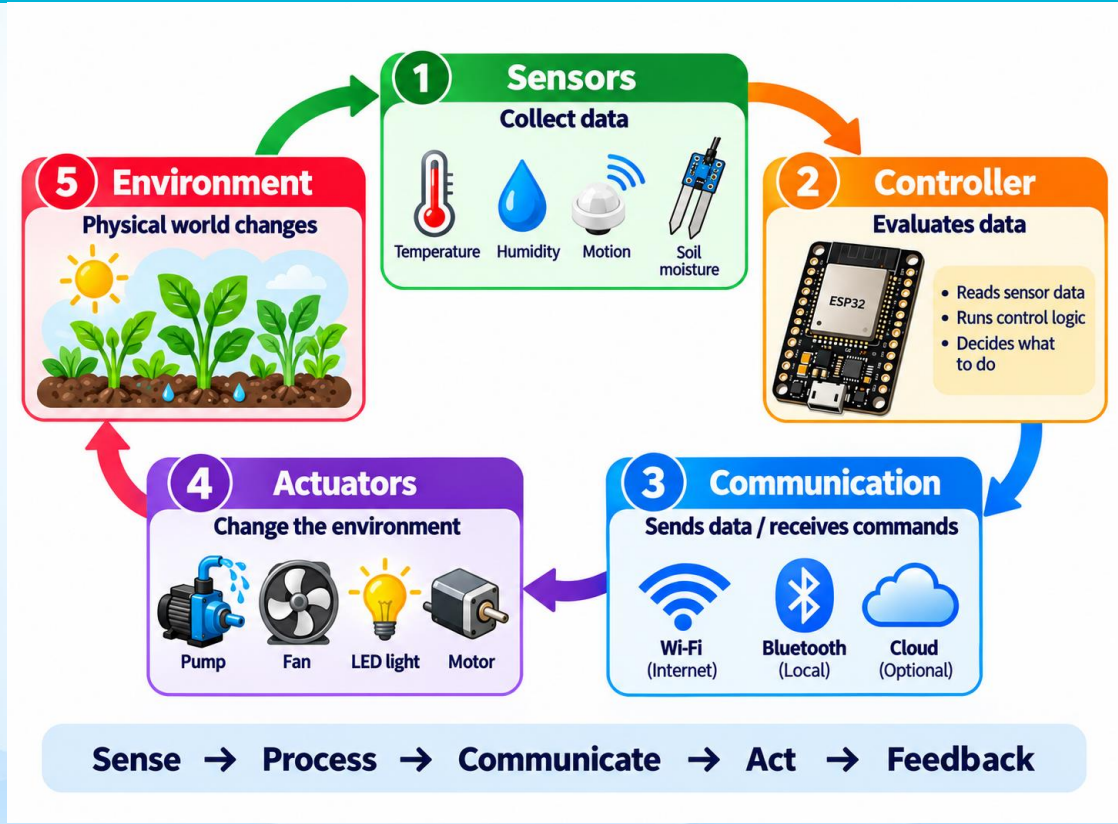
Complete IoT Architecture

Data moves upward, control commands can move downward



The IoT Control Loop

A repeating cycle links sensing and action



Sensors collect data → the controller evaluates it → the device communicates → actuators change the environment.



Summary

The essential model to remember

SENSE → **PROCESS** → **COMMUNICATE** → **ACT**

Sensors

Provide input from the physical world.

Microcontrollers

Process data and execute control logic.

Actuators

Create physical output in response to commands.

Foundation of practical IoT systems

Summary

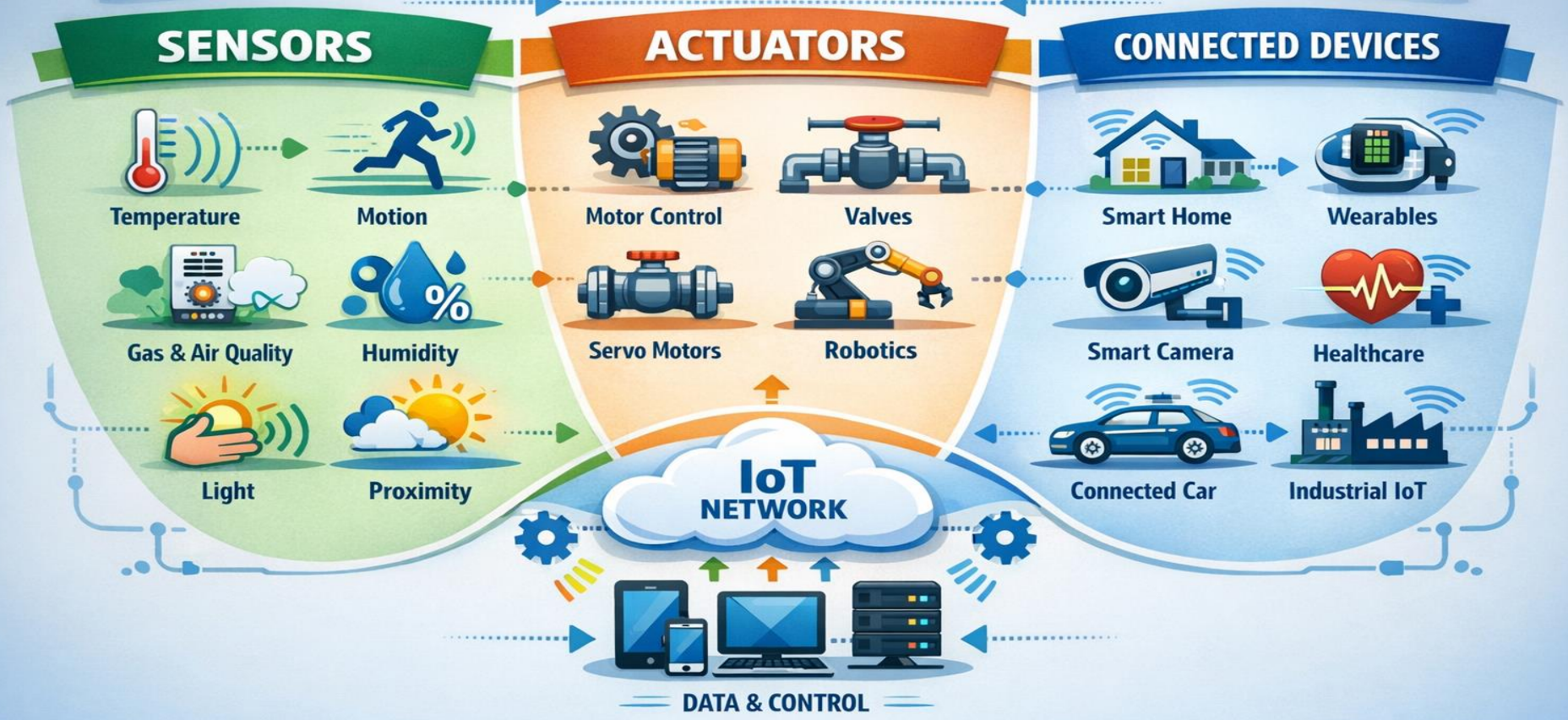
The fundamental hardware model of an IoT device.



Sensors provide inputs. Microcontrollers process and control. Actuators provide outputs.

SENSE → PROCESS → COMMUNICATE → ACT

Sensors, Actuators & Devices in IoT



IOT sensors simulator

Wokwi is an online Electronics simulator. You can use it to simulate Arduino, ESP32, STM32, and many other popular boards, parts and sensors.

https://wokwi.com/

https://docs.wokwi.com/?utm_source=wokwi

Practice exercise:1

A smart home system automatically controls room temperature. It contains:

- A temperature sensor.
- An ESP32 microcontroller.
- A Wi-Fi communication module.
- A relay connected to an air conditioner.
- A smartphone application.

Questions:

1. Identify the IoT device in this system.
2. Which component measures the temperature?
3. Which component processes the collected data?
4. Which component performs the physical action?
5. What is the role of Wi-Fi?
6. Draw the architecture and show the direction of data and control commands.

Practice exercise:2

For each component, classify it as a Sensor, Actuator, or Both.
Then answer:
1.What is the difference between a sensor and an actuator?
2.Can an IoT device contain both sensors and actuators? Give an example.

Component	Classification
Temperature sensor	?
Electric motor	?
LED	?
Ultrasonic sensor	?
Servo motor	?
Smart door lock with position feedback	?

Practice exercise:3

A student wants to develop three IoT applications:

- Application A: A simple temperature monitoring system with very low power consumption.
- Application B: A smart home system requiring Wi-Fi and Bluetooth.
- Application C: An image-processing system connected to a camera.

Choose the most appropriate platform from:

- Arduino Uno
- ESP32
- Raspberry Pi

Complete the table:

Application	Selected platform	Justification
A	?	?
B	?	?
C	?	?

Practice exercise:4

For each application, choose the most appropriate technology from Wi-Fi, Bluetooth Low Energy (BLE), Zigbee, LoRaWAN, and GSM/4G. Justify your choices.

Scenario	Requirement	Your choice
A	Smart watch communicating with a smartphone	?
B	Agricultural sensor transmitting data over 5 km with a 3-year battery target	?
C	Smart home network connecting 20 sensors	?
D	Security camera transmitting video to the Internet	?
E	Remote water meter sending readings through cellular coverage	?

Thank you for your attention

